

General Chemistry (CH105) Midterm Exam

Nov. 13th, 2022, 10:00–12:00 am

Section A: Multiple Choice – Select the one that is best in each case. (40pts)
Number per questions: 20 Point of question: 2

A1) Which one of the following is often easily separated into its components by simple techniques such as filtering or decanting?

- A) heterogeneous mixture B) compounds C) homogeneous mixture
D) elements E) solutions

A2) Which property of the following is extensive property?

- A) temperature B) mass C) density
D) melting point E) hardness

A3) In the following list, only _____ is chemical change.

- A) a wet towel dries in the sun B) dissolving sugar in water
C) the density of gold is 19.3 g/mL
D) an antacid tablet decreases the acidity of an over acid stomach
E) compression of oxygen gas

A4) Which of the following has the same number of significant figures as the number 1.00310?

- A) 1×10^6 B) 5.119 C) 199.791
D) 100 E) 8.66

A5) Which statement below is incorrect?

- A) Of the three types of radioactivity characterized by Rutherford only gamma ray shows no response to an electric field.
B) Atoms are composed of protons, neutrons, and electrons in Dalton's atomic theory.
C) The gold foil experiment performed in Rutherford's lab led to the discovery of the atomic nucleus.
D) The charge on an electron was determined in the Millikan oil drop experiment.
E) Cathode rays travel in straight lines in the absence of electric or magnetic fields.

A6) An elemental ion has 22 protons, 20 electrons, and 26 neutrons. Which isotope is it?

- A) $^{47}_{26}\text{Fe}$ B) $^{48}_{26}\text{Fe}$ C) $^{48}_{22}\text{Ti}$
D) $^{48}_{22}\text{Ti}^{2+}$ E) $^{48}_{22}\text{Ti}^{2-}$

A7) The elements in groups 1A, 6A, and 7A are called

- A) alkaline earth metals, halogens, and chalcogens
B) alkali metals, chalcogens, and halogens
C) alkali metals, halogens, and noble gases
D) alkaline earth metals, transition metals, and halogens
E) halogens, transition metals, and alkali metals

A8) Which of the following statements is true about the following list of compounds?

N_2O , Na_2O , KCl , SiF_4 ,

- A) N_2O is the only molecular compound.
B) Na_2O , KCl and SiF_4 are the only ionic compounds.
C) KCl is the only ionic compound.
D) N_2O , Na_2O and SiF_4 are the only molecular compounds.
E) N_2O and SiF_4 are the only molecular compounds.

A9) Element X has three naturally occurring isotopes. The masses (amu) and % abundances of the isotopes are given in the table below. The average atomic mass of the element is _____ amu.

Isotope	Abundance	Mass
^{38}X	5.07	37.919
^{39}X	15.35	39.017
^{42}X	79.85	42.111

- A) 41.54 B) 39.68 C) 39.07
D) 38.64 E) 33.33

A10) Consider the following four formulas: SO_2 , B_2H_6 , CH , $\text{C}_4\text{H}_2\text{O}_2$. Which of these formulas could be both a molecular and an empirical formula?

- A) SO_2 B) B_2H_6 C) CH
D) $\text{C}_4\text{H}_2\text{O}_2$ E) SO_2 , B_2H_6

A11) Indicate the combination reaction of the following equations.

- A) $\text{PbCO}_3(\text{s}) \rightarrow \text{PbO}(\text{s}) + \text{CO}_2(\text{g})$
B) $2\text{HNO}_3(\text{aq}) + \text{Sr}(\text{OH})_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{Sr}(\text{NO}_3)_2(\text{aq})$
C) $3\text{Mg}(\text{s}) + \text{N}_2(\text{g}) \rightarrow \text{Mg}_3\text{N}_2(\text{s})$
D) $\text{C}_7\text{H}_8\text{O}_2(\text{l}) + 8\text{O}_2(\text{g}) \rightarrow 7\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{g})$
E) $2\text{KCl}(\text{aq}) + \text{Pb}(\text{NO}_3)_2(\text{aq}) \rightarrow \text{PbCl}_2(\text{s}) + 2\text{KNO}_3(\text{aq})$

A12) A 6.25×10^{-3} mol sample of aluminum nitrate ($\text{Al}(\text{NO}_3)_3$) contains _____ atoms of oxygen.

- A) 1.12×10^{22} B) 3.39×10^{22} C) 3.76×10^{22}
D) 1.33 E) 7.21×10^{24}

A13) The formula weight of potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) is _____ amu.

- A) 107.09 B) 255.08 C) 242.18
D) 294.18 E) 333.08

A14) Based on the activity series, which one of the reactions below will occur?

- A) $\text{Zn}(\text{s}) + \text{MnI}_2(\text{aq}) \rightarrow \text{ZnI}_2(\text{aq}) + \text{Mn}(\text{s})$
B) $\text{SnCl}_2(\text{aq}) + \text{Cu}(\text{s}) \rightarrow \text{Sn}(\text{s}) + \text{CuCl}_2(\text{aq})$
C) $2\text{AgNO}_3(\text{aq}) + \text{Pb}(\text{s}) \rightarrow 2\text{Ag}(\text{s}) + \text{Pb}(\text{NO}_3)_2(\text{aq})$
D) $3\text{Hg}(\text{l}) + 2\text{Cr}(\text{NO}_3)_3(\text{aq}) \rightarrow 3\text{Hg}(\text{NO}_3)_2(\text{aq}) + 2\text{Cr}(\text{s})$
E) $3\text{FeBr}_2(\text{aq}) + 2\text{Au}(\text{s}) \rightarrow 3\text{Fe}(\text{s}) + 2\text{AuBr}_3(\text{aq})$

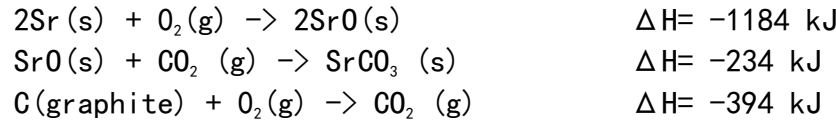
A15) Which compound has the atom with the highest oxidation number?

- A) CaS B) Na_3N C) MgSO_3
D) $\text{Al}(\text{NO}_2)_3$ E) NH_4Cl

A16) The net ionic equation for formation of an aqueous solution of NiI_2 accompanied by evolution of CO_2 gas via mixing solid NiCO_3 ; and aqueous hydriodic acid is _____

- A) $2\text{NiCO}_3(\text{s}) + \text{HI}(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g}) + 2\text{Ni}^{2+}(\text{aq})$
B) $\text{NiCO}_3(\text{s}) + \text{I}^-(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g}) + \text{Ni}^{2+}(\text{aq}) + \text{HI}(\text{aq})$
C) $\text{NiCO}_3(\text{s}) + 2\text{H}^+(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g}) + \text{Ni}^{2+}(\text{aq})$
D) $\text{NiCO}_3(\text{s}) + 2\text{HI}(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g}) + \text{NiI}_2(\text{aq})$
E) $\text{NiCO}_3(\text{s}) + 2\text{HI}(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g}) + \text{Ni}^{2+}(\text{aq}) + 2\text{I}^-(\text{aq})$

A17) Calculate the standard enthalpy of formation of solid SrCO_3 , given the following data:



- A) -826 kJ/mol B) -628 kJ/mol C) -234 kJ/mol
D) -1812 kJ/mol E) -1220 kJ/mol

A18) According to the first law of thermodynamics, which following is false?

- A) Energy is conserved during any process.
B) $\Delta E = q + w$.

- C) Any energy that is lost by a system must be gained by the surroundings, and vice versa.
 D) The amount of work done during a change is independent of the pathway of that change.
 E) Energy can be neither created nor destroyed.

A19) What are the possible values of m_l for 4p subshell?

- A) 0, 1
 B) -1, 0, 1
 C) 1, 2, 3
 D) -1, 0
 E) 1, 2, 3, 4

A20) What is the de Broglie wavelength (m) of a 25 g object moving at a speed of 5.0 m/s?

- A) 1.9×10^{32} m
 B) 5.3×10^{-33} m
 C) 6.6×10^{-36} m
 D) 3.32×10^{-36} m
 E) 3.02×10^{15} m

Section B: Short Answer – Write legibly and show all work for all steps in the problem. (24Pts)

Number of questions: 6

Points per question: 4

B1) A sample of 70.5 mg of potassium phosphate is added to 15.0 mL of 0.050 M silver nitrate, resulting in the formation of 89.8 mg precipitate.

- (a) What is the limiting reactant in the reaction?
 (b) Calculate the percent yield, of the precipitate that forms.

B2) Using only the periodic table, arrange the ions or molecules in order of

- (a) increasing radius: Be^{2+} , Ca^{2+} , Mg^{2+} ;
 (b) decreasing acidity when dissolved in water: CO_2 , CaO , SiO_2 , N_2O_5 .

B3) The following electron configurations represent excited states. Identify the element and indicate the number of unpaired electrons in its ground-state condensed electron configuration.

- (a) $1s^2 2s^2 2p^2 3s^2$, (b) $[\text{Ar}] 4s^1 3d^{10} 4p^2 5p^1$. (c) $[\text{Kr}] 5s^2 4d^2 5p^1$, (d) $[\text{Kr}] 4d^5 5p^1$.

B4) Imagine a hypothetical process in which the methane molecule is "expanded" by simultaneously extending all four C–H bonds to infinity. We then have the process $\text{CH}_4(\text{g}) \rightarrow \text{C}(\text{g}) + 4 \text{H}(\text{g})$.

(a) Compare this process with the reverse of the reaction that represents the standard enthalpy of formation of $\text{CH}_4(\text{g})$. Calculate the enthalpy change in each case.

(b) Which is the more endothermic process?

Substance	C (s, diamond)	C (s, graphite)	C(g)	$\text{CH}_4(\text{g})$	H(g)	$\text{H}_2(\text{g})$
ΔH°_f (KJ/mol)	1.88	0	718.4	-74.8	217.94	0

B5) (a) Compared Li and Na, which has larger first ionization energy? And explain why.

(b) Does magnesium have positive or negative electron affinity? And explain why.

B6) Give name for the following compounds: (a) $\text{Al}(\text{OH})_3$, (b) H_2SO_3 , (c) P_4S_6 , (d) $\text{Fe}_2(\text{CO}_3)_3$.

Section C: Long Answer – Write legibly and show all work for all steps in the problem. For mathematical relationships show both the equations used and substitution of values into the equation. Maintain correct units throughout your calculations and report your answers with the correct significant figures. (36 Pts)

Number of questions: 4

Points per question: 9

C1) A stock solution of NH_3 is 14.8 M.

- (a) How many milliliters of stock solution should you dilute to make 1000.0 mL of 0.250 M NH_3 ?
 (b) If you take a 10.0 mL portion of the stock solution and dilute it to a total volume of 0.500 L, what will be the concentration of the final solution?
 (c) A basic solution is prepared by dissolving 12.50 g of $\text{Sr}(\text{OH})_2$ in water to make 50.00 mL of solution. then the basic solution prepared is used to titrate HNO_3 solution of unknown

concentration. Write a balanced molecular equation to represent the neutralization reaction.
 (d) If 23.9 mL of the basic solution was needed to neutralize a 37.5 mL HNO_3 solution, what is the concentration (molarity) of the acid?

C2) Titanium metal requires a photon with a minimum energy of 6.94×10^{-19} J to emit electrons.

- (a) What is the wavelength of this light?
 (b) Is it possible to eject electrons from titanium metal using visible light, which wavelength ranges from 400–750 nm? Explain why.
 (c) If titanium is irradiated with light of wavelength 263 nm. what is the maximum possible Kinetic energy of the emitted electrons?
 (d) What is the maximum number of electrons that can be freed by a burst of light whose total energy is 2.30 MJ?

C3) A coffee-cup calorimeter contains 150.0 g of water at 25.1 °C. A 121.0-g block of copper metal is heated to 100.4 °C by putting it in a beaker of boiling water. The specific heat of $\text{Cu}(\text{s})$ is 0.385 J/g·K. The Cu is added to the calorimeter, and after a time the contents of the cup reach a constant temperature of 30.1 °C.

- (a) Determine the amount of heat, in J, lost by the copper block.
 (b) Determine the amount of heat gained by the water. The specific heat of water is 4.18 J/g·K.
 (c) The difference between your answers for (a) and (b) is due to heat loss through the Styrofoam cups and the heat necessary to raise the temperature of the inner wall of the apparatus. The heat capacity of the calorimeter is the amount of heat necessary to raise the temperature of the apparatus (the cups and the stopper) by 1 K. Calculate the heat capacity of the calorimeter in J/K.
 (d) What would be the final temperature of the system if all the heat lost by the copper block were absorbed by the water in the calorimeter?

C4) Consider the first ionization energy of neon and the electron affinity of fluorine.

- (a) Write equations, including electron configurations, for each process.
 (b) These two quantities have opposite signs. Which will be positive, and which will be negative?
 (c) Would you expect the magnitudes of these two quantities to be equal? If not, which one would you expect to be larger? And explain why.